

Examiner reports

Q2.

In part (a)(i), the partial fractions were found correctly by most students, substituting $x = 0$ and $x = 3$ being preferred over setting up simultaneous equations or a mix of both.

Most students went on to integrate the partial fraction form correctly, although some omitted the brackets on $\ln(x - 3)$, but this was condoned. Some students were confused by the integral of $\frac{1}{x}$, with some trying to make this become x° . Very few students produced complete nonsense for both integrals, although some differentiated.

Students attempted part (b)(i) by long division or equating coefficients and were usually successful in correctly finding all three coefficients. Those who chose equating coefficients often did far more work than was necessary, as they multiplied out and regrouped the

whole expression and were more likely to make an error. Relatively few used $x = -\frac{1}{2}$ to find r first, and those who did rarely progressed beyond a value for r .

In part (b)(ii), although most students attempted to use their result from part (b)(i), most of these failed to realise that r was not multiplied by $(2x - 1)$, and simply subtracted r from q to give a relatively simple polynomial integral, which gained no credit. Those who did use algebra correctly to set up the integral, usually gave the integral as a log term, but often with an error in the coefficient. Here too, some students differentiated rather than integrated.

Q3.

Part (a) was very well answered, with the majority of candidates obtaining the correct function. The main error made by candidates who obtained a $\ln$ function was with the multiplying constant, which was often 2 or was completely missing. Omission of brackets was also a problem.

Not many of the candidates answered part (b) fully correctly. Only a few integrated by parts 'the

wrong way round' and some weaker candidates put $u = x$ and $v = \sin \frac{x}{2}$. However most knew

that they should integrate the $\sin \frac{x}{2}$ term, but did it wrongly, getting $\pm \cos \frac{x}{2}$ or more commonly

$\pm \frac{1}{2} \cos \frac{x}{2}$. If the first method mark was earned, the second was generally scored as well.

Q4.

In part (a) a repeated term partial fractions expression such as this is the most complex form in the specification, but most candidates approached it with confidence and demonstrated they knew how to find the coefficients. Most chose to use values of x to find

C and A in that order, and then either a third value of x , or equated coefficients in order to find a value for B . This approach was generally more successful than those who equated coefficients to set up and solve three simultaneous equations. Most had the value of C correct, and similarly A although

$\frac{1}{4}A = 1$ often resulted in $A = \frac{1}{4}$. However, about 60 % of the candidates had all of A , B and C correct.

In (b) many candidates ignored the partial fractions they had just found, and attempted all sorts of nonsensical ways of “integrating” the right hand side, some before even attempting to separate the variables, using what appeared to be an attempt to use parts. Some who did separate the y term to the left hand side, then attempted to multiply out the right hand side and integrate term by term, taking no notice of denominators. About 40% of the candidates gained no credit in part (b). However, many did know the first step in solving the differential equation was to separate the variables and use the partial fractions. The notation was often poor as was the

algebra in handling the $2\sqrt{y}$, the latter often appearing as such on the left hand side and not as its reciprocal. Many mistakes were made in attempting to integrate the various terms. Indices

caused problems for many with the term $\frac{1}{\sqrt{y}}$ being expressed as both $y^{\frac{1}{2}}$ and $y^{-\frac{1}{2}}$. Credit was

given for integrating $\frac{k}{\sqrt{y}}$ to $2k\sqrt{y}$, for any k . In the two log integrals on the right-hand side coefficient and sign errors were common and many thought the third term had a

logarithmic integral as well. Others assumed this term integrated to $\frac{x}{1-x}$ as given in the answer. Some candidates did integrate this term correctly by inspection, whereas others used substitution, also correctly. Most candidates included a constant and tried to find it using the boundary conditions given, but they were only credited with the correct value if it was exact, and found from a correct expression. The penultimate mark was for correctly combining the logarithm terms in a candidate’s solution; many failed to do this correctly, multiplying terms when they should have been dividing or vice versa depending on signs. Those who had made errors and so didn’t have all the coefficients equal to 2, did make some valiant efforts using powers to combine their terms, and got the credit if it was done correctly. Everything had to be correct to gain the last mark, and this was achieved by rather less than 5% of the candidates.

Q5.

In part (a), most candidates knew to multiply through by the denominator and many did this correctly and went on to find A and B successfully. However, some candidates did not apparently know how to handle the 2, some omitting it altogether and some including it but not multiplying it by the denominator. Some candidates multiplied out the denominator and divided by the resulting quadratic expression to separate out the 2, then worked with the resulting algebraic fraction. Although this was unnecessary, it was a perfectly acceptable method. Relatively few candidates used the method of setting up simultaneous equations to find A and B .

In part (b), virtually all candidates knew they were to use the partial fractions to do the integration, and most recognised that log integrals were involved. The common error was

to miss the multiplier $\frac{1}{5}$ when integrating $(5x - 1)^{-1}$. Several candidates inverted the

coefficient, giving it as $\frac{5}{7}$. Although most candidates integrated 2 to $2x$, many omitted the arbitrary constant.

Q6.

- (a) (i) The vast majority of candidates found the partial fractions correctly, most substituting the two expected values of x . Some proceeded by setting up and solving simultaneous equations and some a hybrid of both methods, but few errors were made. A few lost all the marks here through having an algebraically incorrect opening line.
- (ii) Here too, most candidates knew these were log integrals with relatively few showing no knowledge of this. The common error was to omit the $\frac{1}{3}$ and a few candidates didn't have their log expressions in brackets and were penalised.
- (b) Various attempts at this question were seen, ranging from no attempt to a time consuming approach involving setting up simultaneous equations in P , Q and R following substitution of values of x , usually 0, 1 and 2. Most candidates who attempted this either set up just two equations, and thus found it was insoluble, or simply made errors in the algebra. The most successful and efficient method was algebraic long division. Relatively few candidates showed any knowledge of the technique of manipulating the numerator, as anticipated in the mark scheme. Some attempted to factorise either the numerator or denominator, or both, and made no further progress because these expressions did not factorise.

Q7.

Part (a) was generally done very well with most candidates gaining full marks, although many methods were seen in the finding of A and B , ranging from inspection to division to rearranging the numerator. Most candidates also integrated correctly, although many omitted the constant of integration and were penalised.

In part (b), most candidates had the opening line of clearing fractions correct, although a few had an extra $(5 - x)$ and could score no further marks. The finding of P , Q and R was done accurately and concisely by many candidates. Such candidates had usually

substituted 5 , $-\frac{1}{3}$ and 0 for x .

The occasionally-seen error was in finding the value of Q . Those candidates who proceeded by multiplying out and equating coefficients often made numerical and/or algebraic errors and sometimes abandoned their efforts when they saw the unlikely looking numbers emerging. This approach cost them marks and time. Some candidates used a hybrid of both methods.

All candidates who attempted part (b)(ii) knew that they were to use the partial fractions and, with the occasional exception, knew that the first two integrals were logarithmic.

However, many mistakes were made in the coefficient's value and/or sign. Relatively few were successful on the third integral. Many just had the sign incorrect, although some assumed it too was a logarithm, and some expanded $(5 - x)^2$ into a quadratic expression and ended up going nowhere with it.

Q8.

Part (a)(i) was well answered.

Candidates who saw the connection with part (a)(i), usually made a very good attempt at part (a)(ii), often with complete success. Limits did cause a problem where candidates had not bracketed $2x^2 - 8x + 3$, and some candidates did not leave the answer in the required

form but left their answer as $\frac{1}{4} \ln 9$.

Part (b) was well answered by candidates who were able to write the integral in terms of u successfully. Many candidates lost a mark through omission of either du or c . A disturbing

number of candidates thought that $\frac{1}{3} \times \frac{1}{3} = \frac{1}{6}$. Some clearly able candidates lost the last mark by spoiling an otherwise acceptable answer.

Q9.

This question, surprisingly, was not done well. Part (a) was intended as a straight forward partial fractions question, although some candidates did not see it that way. Many apparently guessed at the value of k , 3 being a common wrong answer. Some candidates replaced the k with the conventional A and B , and then demonstrated that $A = B$. Many of these, having arrived correctly at $6A = 3$, then deduced that $A = 2$. This 2, or other values,

often became $\frac{1}{2}$ in part (b) without any supporting evidence and so lost further marks. Very few candidates in this situation seemed to have checked for a mistake, despite the $\frac{1}{2}$ being given in part (b).

In part (b), most candidates picked up a mark for recognising a log integral, although the

coefficients were often incorrect, particularly the sign in integrating the $\frac{1}{3-x}$ term. However, most candidates could evaluate their log expression correctly using the laws of logs.

Q10.

Nearly all candidates successfully found the partial fractions in part (a), most substituting $x = 1$ and $x = -1$; although some used other values or equated coefficients.

In part (b) most candidates then went on to use their partial fractions to integrate the given expression correctly. The error that was sometimes seen was to integrate $(x - 1)^{-1}$ to $(x - 1)^0$, and similarly for $(x + 1)^{-1}$.

Most candidates could make a start to part (c) of the question, but many rearranged the 2 and 3, and this detracted from the realisation that with y and 3 taken to the other side of the equation, they had the integral they had just done in part (b). Some who did realise this, proceeded to give an incorrect integral because they had “lost” the 2. Most candidates knew they needed to separate the variables but for many candidates the algebra and ensuing attempts at integration were full of errors. Turning the denominators y and $(x^2 - 1)$ into numerators was common, as was integrating $(x^2 - 1)^{-1}$ to an expression involving $\ln(x^2 - 1)$.

Those who had taken the 2 to the left hand side, often got confused with the integral of $(2y)^{-1}$, this becoming $\ln 2y$. The problem for most candidates seemed to be that they associated the 2 in the question with the 2 in the answer, although the 2 in the answer came from the given values of x and y . Most candidates did include an arbitrary constant and tried to use the given values to find it, often claiming, after some dubious manipulation, that they had derived the given result. Few fully correct answers were seen, but where they were, it was reassuring to see confident use of the log laws in deriving the result.

Q11.

This question was generally done well with many candidates gaining full marks. Marks tended to be lost in part (a) rather than part (b), possibly because the latter was a more conventional partial fractions problem. There were several different approaches to part (a) including manipulating the numerator to contain $A(x - 3)$, algebraic division and cross multiplying and using partial fraction type techniques, or some mix of these. Errors tended to be algebraic or numerical. In part (a)(ii) it was notable that some candidates did not attempt a logarithmic integral, although they went on to do so in part (b).

In part (b) virtually all candidates scored well, either using the expected two values of x or equating coefficients to find the values of P and Q . Again a few made algebraic or numerical errors, with very few cross multiplying incorrectly. Similarly virtually all candidates went on to give some form of logarithmic integrals, sometimes with errors in the coefficients. Very few non-sensible attempts at the integral were seen.

Q12.

- (a) (i) This was reasonably well done, although some candidates integrated.
- (ii) Many candidates failed to appreciate the significance of part (a)(i) and obtained complicated incorrect expressions. Those candidates who did appreciate the relevance of part (i) either obtained the correct answer or $2\ln|x^4 + 2x|$.
- (b) (i) This was fairly well answered. However, many candidates lost marks because they did not realise that in any substitution question, apart from
- $$\frac{du}{dx}$$
- replacing functions of x with functions of u , they have to find and use
- (ii) The integration was very well answered by the majority of candidates with many accurately completing the question. Problems did arise through the use of incorrect limits. Many candidates converted their integration answers in u to terms in x and then used the original limits. This method was equally

successful.

Q13.

There were many good answers to this question, with full or nearly full marks.

Part (a) Partial fractions is an area that is generally known well, although the presence of the 3 led some candidates to make an error, some ignoring it altogether, and others not multiplying it by the common denominator. Some candidates simplified the given expression by dividing it out, which was perfectly valid. Those candidates who substituted $x = 1$ and $x = -$ usually found the coefficients correctly; many candidates who chose to set up simultaneous equations by equating coefficients made an error.

Part (b) Virtually all candidates knew they were to use the partial fractions to do the integral and that it involved $\ln(3x - 1)$ and $\ln(x - 1)$, and very few nonsense “integrations” were seen. The common errors were in the coefficients with the $1/3$ from $3x$ being missed, or the coefficients given as fractions $1/6$ and $1/4$.



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